

Digital Circuits [ECE 111] End Semester Examination 2024

All questions are compulsory.

Section A: Objective

1. Consider the following set 0, 1, 2, ..., 512. How many bits are needed to represent the elements of this set using a binary system? [1 Mark]
(a) 8
(b) 9
(c) 10
(d) 11
2. Convert $(214)_8$ into decimal. [1 Mark]
(a) $(139)_{10}$
(b) $(140)_{10}$
(c) $(141)_{10}$
(d) $(142)_{10}$

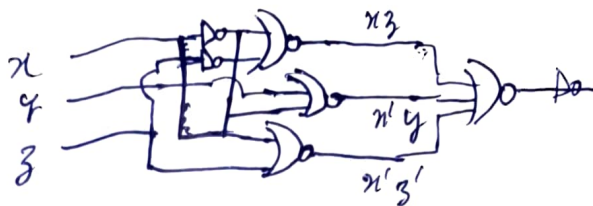
Handwritten work:
 $2 \times 8^2 + 1 \times 8^1 + 4 \times 8^0 = 128 + 8 + 4 = 140$
3. Add $(+38)_{10}$ and $(-20)_{10}$ using 2's complement system, we get which of the following number (neglect the overflow bit if any) [1 Mark]
(a) 11011001
(b) 0010010
(c) 100000110
(d) 110110011
4. Complement the expression $A'B + CD'$. Choose the correct answer from the following options. [1 Mark]
(a) $(A' + B)(C' + D)$
(b) $(A + B')(C' + D)$
(c) $(A' + B)(C' + D)$
(d) $(A + B')(C + D')$
5. How many minimum numbers of NOR-gates are required to implement a two input OR function? [1 Mark]
(a) 2
(b) 3
(c) 4
(d) 5
6. How many gates are required to implement a 2×1 Multiplexer? [1 Mark]
(a) 2
(b) 3
(c) 4
(d) 5

[Turn over

7. In a NAND gated S-R latch, when $SR=00$, then output will be.

[1 Mark]

- (a) Invalid
- (b) SET
- (c) RESET
- (d) No Change



Section B: Not Objective

1. Implement the Boolean Function $F = xz + x'z' + x'y$ with NOR and inverter gates. [2 Mark]
2. I want to compare two 4 bit signed numbers based on a 2's complement representation. Use a "subtractor" circuit to come up with a magnitude comparator. [2 Mark]
3. Consider an SR latch implemented using two NOR gates. Initially, both inputs $S = 0$ and $R = 0$. The latch is in a stable state with $Q = 1$ and $\bar{Q} = 0$. Now, the input S is momentarily set to 1 and then returned to 0. What will be the final state of Q and $\bar{Q}$? [2 Mark]
4. Consider the table below of a sequential circuit with current state (c.s), input (inp) and new state (n.s) as the three columns. Implement the state table via a T flip flop based circuit. [3 Mark]

c.s	inp	n.s
0	1	1
1	1	0
0	0	1
1	0	1

Table 1: Que

5. A digital system has a clock generator that produces pulses at a frequency of 80 MHz. Design a circuit that provides a clock time of 50 ns. [3 Mark]
6. Draw a circuit using three T flip flops that does reverse counting from 7 to 1. Explain the way you reached the circuit by writing appropriate steps along the way. [3 Mark]
7. Here is a K-MAP table for four variables. [2 Mark]

$AB \backslash CD$	00	01	11	10
00	0	1	0	1
01	1	0	1	0
11	0	1	0	1
10	1	0	1	0

Find the simplified Boolean expression.

8. Simplify the following Boolean expression: $x'yz + xz$ [2 Mark]
9. Convert $(12.375)_{10}$ to the corresponding Hexa-decimal number. [2 Mark]
10. For the circuit given, draw the state transition diagram. [2 Mark]
11. Draw the output timing diagram for a positive edge-triggered J-K flip flop with the given timing diagram of inputs J and K. (Consider SET and RESET signals are in low state). [3 Mark]

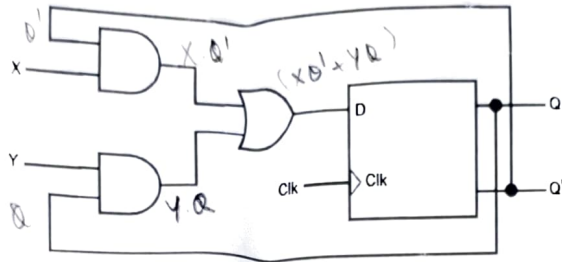


Figure 1: Question 10

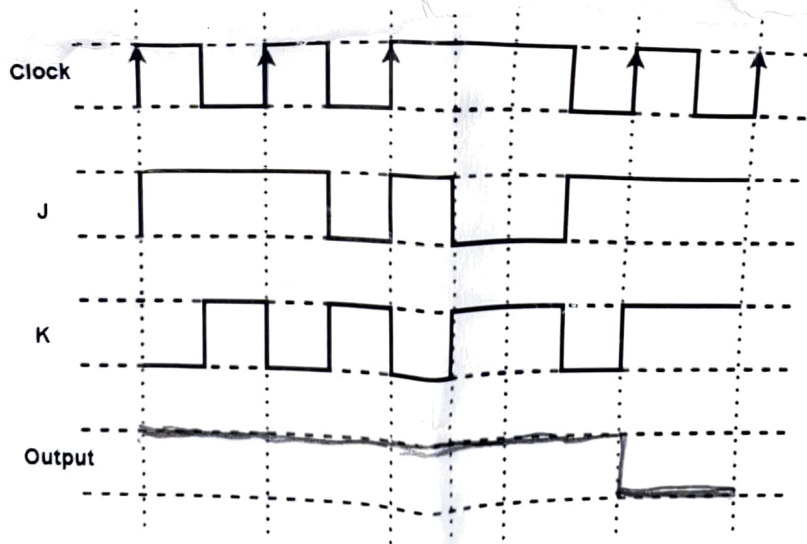


Figure 2: Question 11

J	K	Output
0	0	Q _n
0	1	0
1	0	1